

Post Harvest Spotted Lanternfly Removal

Team: DEAd heAD **Client(s):** Cornell CALS Extension / E&J Gallo Winery / National Grape

Problem Statement: Spotted Lanternflies (SLF) are devastating wineries and grape processors in New York and nearby regions, with losses reaching \$8.8 million in three years. During high-volume harvest periods, post-harvesting facilities must remove SLF quickly, yet current methods continue to miss insects, risking contamination and regulatory noncompliance which can lead to financial and operational strain.

Impact: Solving the SLF contamination would protect the wine supply chain, helping growers in New York preserve crop value while enabling processors to increase throughput, reduce labor, and lower losses, stabilizing supply for distributors and retailers. This keeps regional producers competitive, strengthens the broader agricultural economy, and ensures millions of consumers continued access to quality wine.

Concept A (Primary) - Filtering Brushes: A car wash-like machine where rotors will brush off SLF on the conveyor belt. Brushes will be tuned in stiffness to be able to remove SLF, minimizing product loss.

How it would be used (user flow):

Step 1: Grapes enter the post-processing conveyor belt.

Step 2: Product passes through brush module, where brushes knock SLF off of harvested grapes.

Why it's better than the status quo: This is a mechanically simple mechanism that could prove to be effective, and it's easily integrated into pre-existing post-harvest processing conveyors.

End-of-semester proof-of-concept: A simple motor mechanism with spinning brushes, testing by modeling grapes and SLF with a force gauge to test the force of the brushes.

Concept B (Secondary) - Wind-Sorting Unit: A controlled airflow module that removes lighter SLF from heavier grapes moving along a conveyor belt.

How it would be used (user flow):

Step 1: Grapes enter on a conveyor belt.

Step 2: Belt passes through the wind-sorting unit, where airflow blows SLF off of grape stream.

Why it's better than the status quo: No direct contact with grapes is needed, so grapes will not be harmed during the process. The unit is easy to automate and integrate with existing processing equipment.

End-of-semester proof-of-concept: An airflow unit that uses small air guns or fans to blow SLF off the grapes. Testing by simulating SLF on grapes and using airspeed sensors to measure the force of air.

Key Risks / Unknowns

- For Concept A, there could be clogging due to jamming on the conveyor belt or on the brushes. We will compare with existing designs and test with simulated grapes and SLF.
- For Concept B, airflow intensity and direction will need to be tuned to ensure grape quality does not degrade. We will test this by simulating grape processing under varied conditions.
- Both proposed solutions include the possibility of excessive grape removal or damage. We will test this by experimenting with different brush stiffnesses, rotation speeds, and airflow intensities.

Questions for the Client

1. Are SLF entering post-harvest processing primarily alive and intact, or dead and fragmented within the grape stream/slush?

Decision affected: Determines whether mechanical separation is viable or if removal must occur earlier, since fragmented SLF may be harder to remove.

2. What does a typical processing layout look like, and where would sanitation, space, or workflow constraints limit adding equipment?

Decision affected: Defines integration limits so the design fits operational and regulatory realities, and supports solutions that can scale across facilities without major infrastructure changes.

3. Which forms of product damage are least acceptable (juice release, skin breakage, berry loss, or cluster disruption)? Please rank if possible.

Decision affected: Sets force and contact limits, guiding mechanism selection and tuning so the design minimizes losses within acceptable damage tolerances.

References

- <https://news.cornell.edu/stories/2025/01/spotted-lanternflies-could-cost-nys-grape-industry-millions>